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БЕЛОРУССКИЙ ГОСУДАРСТВЕННЫЙ УНИВЕРСИТЕТ
ИНФОРМАТИКИ И РАДИОЭЛЕКТРОНИКИ

Факультет компьютерных систем и сетей

Кафедра электронных вычислительных машин

Лабораторная работа № 4
по дисциплине ФПрог

Вариант 1

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Цель: изучить использование Spark Scala.

Теоретические сведения:

Программа, выводящая первое встреченные предложение, содержащее не менее трех слов, а после всего теста:

```
val conf = new org.apache.spark.SparkConf()
val lines = spark.read.textFile("c:\\Users\\afili\\Downloads\\pnaptext.txt")
val words = lines.map(line => line.split(" "))
val longestSentence = lines.filter(line => line.split(" ").length >= 3).first()
val longestSentence2 = lines.foreach(line => println(line))
```

Пример работы:

```
scala> val longestSentence = lines.filter(line => line.split(" ").length >= 3).first()
longestSentence: String = "hello to everybody "

scala> val longestSentence2 = lines.foreach(line => println(line))
hello to everybody
this is first attemp in Spark text processing
Be careful with people No all of them symphatize you
longestSentence2: Unit = ()
```

Программа, выводящая слова, не входящие в стоп-список, после выводящая все слова, начинающиеся на 's', после слова, начинающиеся на 's', имеющие 'r' в середине и заканчивающиеся на 'k':

```
val conf = new org.apache.spark.SparkConf()
val lines = spark.read.textFile("c:\\Users\\afili\\Downloads\\pnaptext.txt")
val wordsRDD = lines.flatMap(line => line.split("\\W+"))
val lowercaseWordsRDD = wordsRDD.map(word => word.toLowerCase())
val stopWords = Set("a", "an", "the", "and", "but", "or", "for", "of")
val filteredWordsRDD = lowercaseWordsRDD.filter(word => !
stopWords.contains(word))
filteredWordsRDD.collect().foreach(println)
val filteredWordsRDD2 = wordsRDD.filter(word => word.startsWith("s"))
filteredWordsRDD2.collect().foreach(println)
val pattern = "^s.*r*k$"
val filteredWordsRDD3 = wordsRDD.filter(word => word.matches(pattern))
filteredWordsRDD3.collect().foreach(println)
```

Примеры работы:

```
scala> filteredWordsRDD.collect().foreach(println)
hello
to
everybody
this
is
first
attemp
in
spark
text
processing
be
careful
with
people
no
all
them
sympthatize
you
```

```
scala> val filteredWordsRDD2 = wordsRDD.filter(word => word.startsWith("s"))
filteredWordsRDD2: org.apache.spark.sql.Dataset[String] = [value: string]

scala> filteredWordsRDD2.collect().foreach(println)
sympthatize
```

```
scala> val filteredWordsRDD2 = filteredWordsRDD.filter(word => word.startsWith("s"))
filteredWordsRDD2: org.apache.spark.sql.Dataset[String] = [value: string]

scala> filteredWordsRDD2.collect().foreach(println)
spark
```

Задание:

Создать собственный текстовый файл на английском или немецком языке – 4-5 предложений.

1. Вывести все слова из текстового файла, исключая stop-слова
2. Вывести все слова, содержащие букву t
3. Вывести все слова, заканчивающиеся на ing
4. Вывести все слова вторая буква которых a
5. Вывести все слова, последняя буква которых s
6. Вывести каждое второе слово

```
// Загрузка текстового файла
val textFile = spark.sparkContext.textFile("textscala.txt")

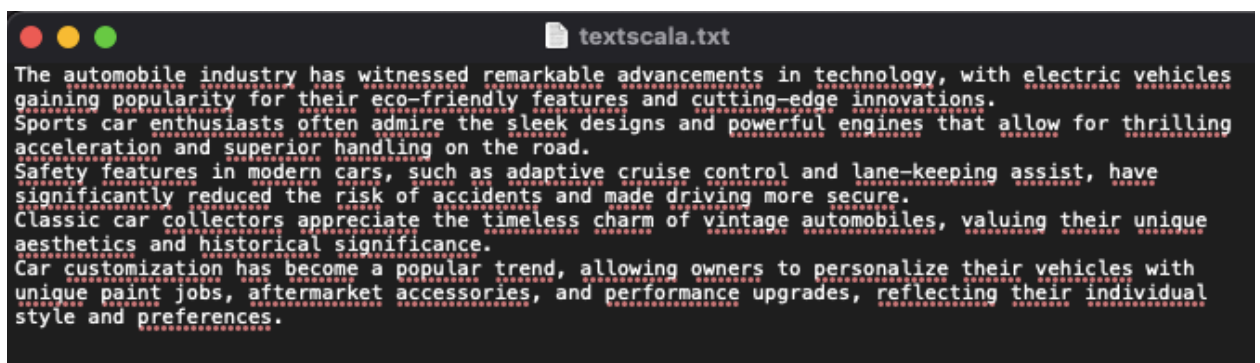
// Определение регулярных выражений
val stopWords = "\\b(the|and|is|in|it|of|to|a)\\b".r
val carOrAvtoPattern = "\\b(car|avto)\\b".r
val myPattern = "\\b.*my\\b".r
val thirdLetterDPattern = "\\b..d.*".r
val threeOrMoreVowelsPattern = "\\b.*[aeiouAEIOU].*[aeiouAEIOU].*[aeiouAEIOU].*\\b".r

// Фильтрация и обработка данных
val words = textFile.flatMap(line => line.split("\\s+"))
val filteredWords = words.filter(word => !
stopWords.pattern.matcher(word.toLowerCase).find)
val carOrAvtoWords = filteredWords.filter(word =>
carOrAvtoPattern.findFirstIn(word.toLowerCase).isDefined)
val myWords = filteredWords.filter(word =>
myPattern.findFirstIn(word.toLowerCase).isDefined)
val thirdLetterDWords = filteredWords.filter(word =>
thirdLetterDPattern.findFirstIn(word.toLowerCase).isDefined)
val threeOrMoreVowelsWords = filteredWords.filter(word =>
threeOrMoreVowelsPattern.findFirstIn(word.toLowerCase).isDefined)

// Вывод результатов
filteredWords.collect().foreach(println)
carOrAvtoWords.collect().foreach(println)
myWords.collect().foreach(println)
thirdLetterDWords.collect().foreach(println)
threeOrMoreVowelsWords.collect().foreach(println)

val secondWord = filteredWords.take(2).last
println(secondWord)
```

Текст в файле:



The automobile industry has witnessed remarkable advancements in technology, with electric vehicles gaining popularity for their eco-friendly features and cutting-edge innovations. Sports car enthusiasts often admire the sleek designs and powerful engines that allow for thrilling acceleration and superior handling on the road. Safety features in modern cars, such as adaptive cruise control and lane-keeping assist, have significantly reduced the risk of accidents and made driving more secure. Classic car collectors appreciate the timeless charm of vintage automobiles, valuing their unique aesthetics and historical significance. Car customization has become a popular trend, allowing owners to personalize their vehicles with unique paint jobs, aftermarket accessories, and performance upgrades, reflecting their individual style and preferences.

Примеры работы:

```

petia — java — spark-shell — 80x52

scala> val textFile = spark.sparkContext.textFile("textscala.txt")
textFile: org.apache.spark.rdd.RDD[String] = textscala.txt MapPartitionsRDD[26]
at textFile at <console>:22

scala>

scala> val stopWords = "\\b(the|and|is|in|it|of|to|a)\\b".r
stopWords: scala.util.matching.Regex = \b(the|and|is|in|it|of|to|a)\b

scala> val carOrAvtoPattern = "\\b(car|avto)\\b".r
carOrAvtoPattern: scala.util.matching.Regex = \b(car|avto)\b

scala> val myPattern = "\\b.*my\\b".r
myPattern: scala.util.matching.Regex = \b.*my\b

scala> val thirdLetterDPattern = "\\b..d.*".r
thirdLetterDPattern: scala.util.matching.Regex = \b..d.*

scala> val threeOrMoreVowelsPattern = "\\b.*[aeiouAEIOU].*[aeiouAEIOU].*[aeiouAEIOU].*\\b".r
threeOrMoreVowelsPattern: scala.util.matching.Regex = \b.*[aeiouAEIOU].*[aeiouAEIOU].*[aeiouAEIOU].*\b

scala>

scala>

scala> val words = textFile.flatMap(line => line.split("\\s+"))
words: org.apache.spark.rdd.RDD[String] = MapPartitionsRDD[27] at flatMap at <console>:23

scala> val filteredWords = words.filter(word => !stopWords.pattern.matcher(word.toLowerCase).find)
filteredWords: org.apache.spark.rdd.RDD[String] = MapPartitionsRDD[28] at filter at <console>:25

scala> val carOrAvtoWords = filteredWords.filter(word => carOrAvtoPattern.findFirstIn(word.toLowerCase).isDefined)
carOrAvtoWords: org.apache.spark.rdd.RDD[String] = MapPartitionsRDD[29] at filter at <console>:25

scala> val myWords = filteredWords.filter(word => myPattern.findFirstIn(word.toLowerCase).isDefined)
myWords: org.apache.spark.rdd.RDD[String] = MapPartitionsRDD[30] at filter at <console>:25

scala> val thirdLetterDWords = filteredWords.filter(word => thirdLetterDPattern.findFirstIn(word.toLowerCase).isDefined)
thirdLetterDWords: org.apache.spark.rdd.RDD[String] = MapPartitionsRDD[31] at filter at <console>:25

```

```
petia — java ◀ spark-shell — 80x52

scala> val threeOrMoreVowelsWords = filteredWords.filter(word => threeOrMoreVowelsPattern.findFirstIn(word.toLowerCase).isDefined)
threeOrMoreVowelsWords: org.apache.spark.rdd.RDD[String] = MapPartitionsRDD[32] at filter at <console>:25

scala>

scala>

scala> filteredWords.collect().foreach(println)
automobile
industry
has
witnessed
remarkable
advancements
technology,
with
electric
vehicles
gaining
popularity
for
their
eco-friendly
features
cutting-edge
innovations.
Sports
car
enthusiasts
often
admire
sleek
designs
powerful
engines
that
allow
for
thrilling
acceleration
superior
handling
on
road.
Safety
features
modern
cars,
such
```

```
petia — java • spark-shell — 80x52
as
adaptive
cruise
control
lane-keeping
assist,
have
significantly
reduced
risk
accidents
made
driving
more
secure.
Classic
car
collectors
appreciate
timeless
charm
vintage
automobiles,
valuing
their
unique
aesthetics
historical
significance.
Car
customization
has
become
popular
trend,
allowing
owners
personalize
their
vehicles
with
unique
paint
jobs,
aftermarket
accessories,
performance
upgrades,
reflecting
their
individual
style
```

```
petia — java ◀ spark-shell — 80x52

preferences.

scala> carOrAvtoWords.collect().foreach(println)
car
car
Car

scala> myWords.collect().foreach(println)

scala> thirdLetterDWords.collect().foreach(println)
industry
cutting-edge
modern
reduced
made
individual

scala> threeOrMoreVowelsWords.collect().foreach(println)
automobile
witnessed
remarkable
advancements
technology,
electric
vehicles
gaining
popularity
eco-friendly
features
cutting-edge
innovations.
enthusiasts
admire
powerful
engines
acceleration
superior
features
adaptive
cruise
lane-keeping
significantly
reduced
accidents
secure.
collectors
appreciate
timeless
vintage
automobiles,
valuing
unique
```



```
aesthetics  
historical  
significance.  
customization  
become  
popular  
allowing  
personalize  
vehicles  
unique  
aftermarket  
accessories,  
performance  
upgrades,  
reflecting  
individual  
preferences.  
  
scala>  
  
scala> val secondWord = filteredWords.take(2).last  
secondWord: String = industry  
  
scala> println(secondWord)  
industry
```